

Surface Metrics Comparison: Human-AI vs Human-Human vs AI-AI

Cross-Condition Text Descriptives on the Exchange-Aligned Structure

PENPAL Project

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1 Overview

This notebook compares surface-level text metrics across conditions using the long-format, exchange-aligned metric tables.

The underlying processed data are now one row per **speaker contribution**, with recovered role metadata:

- **Human / AI** in Human-AI
- **Starter-side / Non-starter-side** in Human-Human and AI-AI

Those role labels are useful for **within-condition description**, but they are not the primary cross-condition comparison axis. For cross-condition comparison, this notebook uses **Slot 1 / Slot 2 asymmetry**.

The notebook therefore no longer assumes old wide `author_1_*` / `author_2_*` columns.

Primary cross-condition object. Per story and per metric m ,

$$A_m(i) = |\overline{m}_{\text{slot}=1, i} - \overline{m}_{\text{slot}=2, i}|.$$

This is the unsigned, label-invariant side-difference magnitude. For each metric we test $A_m \sim \text{condition}$ with ANOVA + Kruskal and apply BH adjustment across the metric panel. The LMM slot-marginal contrasts are retained as supplemental. Starter-mechanism and Human-AI speaker-type decompositions are deferred.

2 Load Data

Table 1: Loaded surface-metric rows by condition

condition	n_stories	n_rows
Human-AI	100	1708
Human-Human	31	558
AI-AI	40	720

3 Metric Selection

Table 2: Surface metrics selected for the comparison notebook

metric_alias	column_name
word_count	n_tokens
sentence_length	sentence_length_mean
lexical_diversity	proportion_unique_tokens
readability	flesch_reading_ease
coherence	first_order_coherence

4 Primary: Per-Story Side-Mean Difference

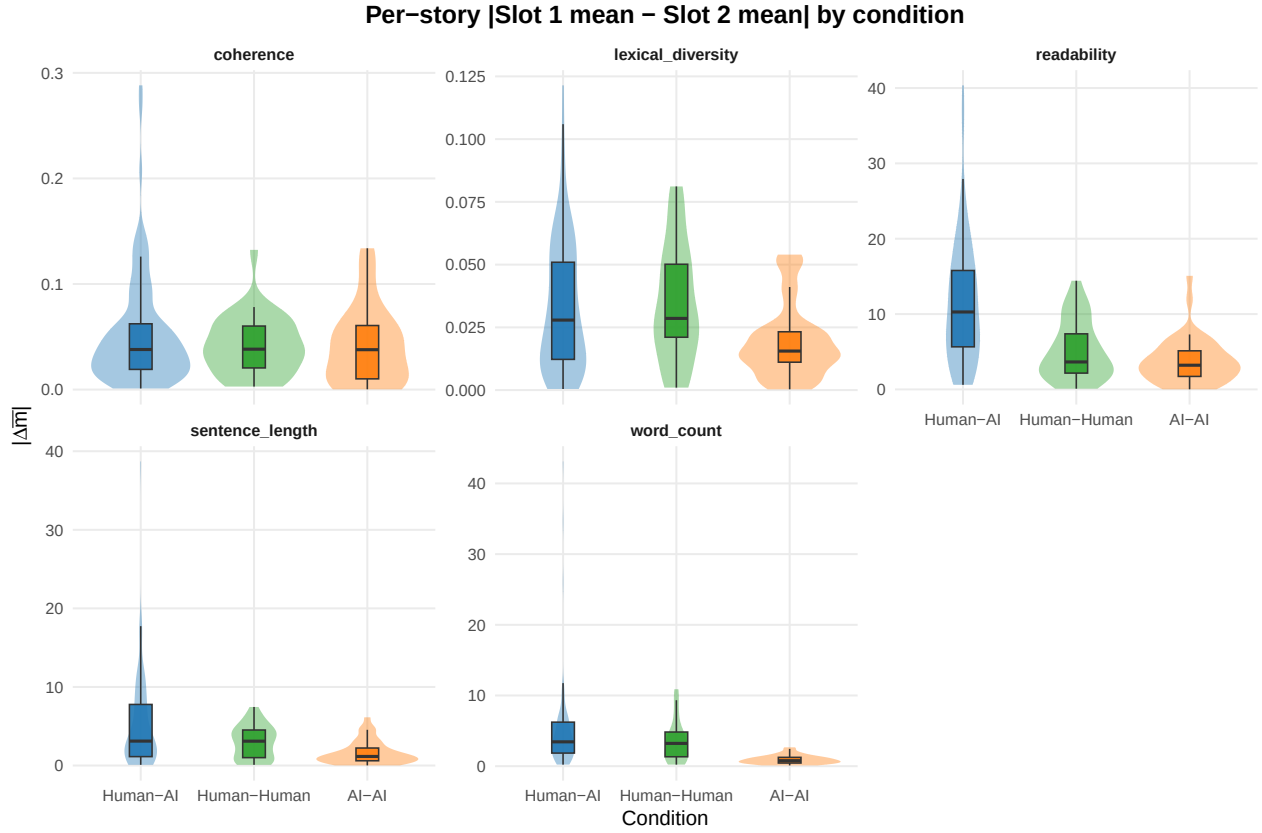
Table 3: Per-story |side-mean difference| by condition, per metric

metric	condition	n	mean_A	median_A	sd_A
coherence	Human-AI	98	0.053	0.038	0.052
coherence	Human-Human	31	0.042	0.038	0.028
coherence	AI-AI	39	0.042	0.038	0.037
lexical_diversity	Human-AI	100	0.033	0.028	0.026
lexical_diversity	Human-Human	31	0.035	0.029	0.023
lexical_diversity	AI-AI	40	0.019	0.016	0.015
readability	Human-AI	100	11.447	10.277	8.084
readability	Human-Human	31	5.159	3.644	3.893
readability	AI-AI	40	3.793	3.210	3.086
sentence_length	Human-AI	100	5.154	3.111	5.658
sentence_length	Human-Human	31	3.023	3.102	2.105
sentence_length	AI-AI	40	1.661	1.157	1.505
word_count	Human-AI	100	4.822	3.444	6.019
word_count	Human-Human	31	3.599	3.222	2.931
word_count	AI-AI	40	0.944	0.778	0.620

Table 4: Primary cross-condition tests for $A_m \sim \text{condition}$, per metric (BH-adjusted across metrics)

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
coherence	0.3126	0.6569	0.3126	0.6569
lexical_diversity	0.0041	0.0060	0.0052	0.0075

metric	anova_p	kw_p	anova_p_BH	kw_p_BH
readability	0.0000	0.0000	0.0000	0.0000
sentence_length	0.0001	0.0001	0.0002	0.0001
word_count	0.0001	0.0000	0.0002	0.0000

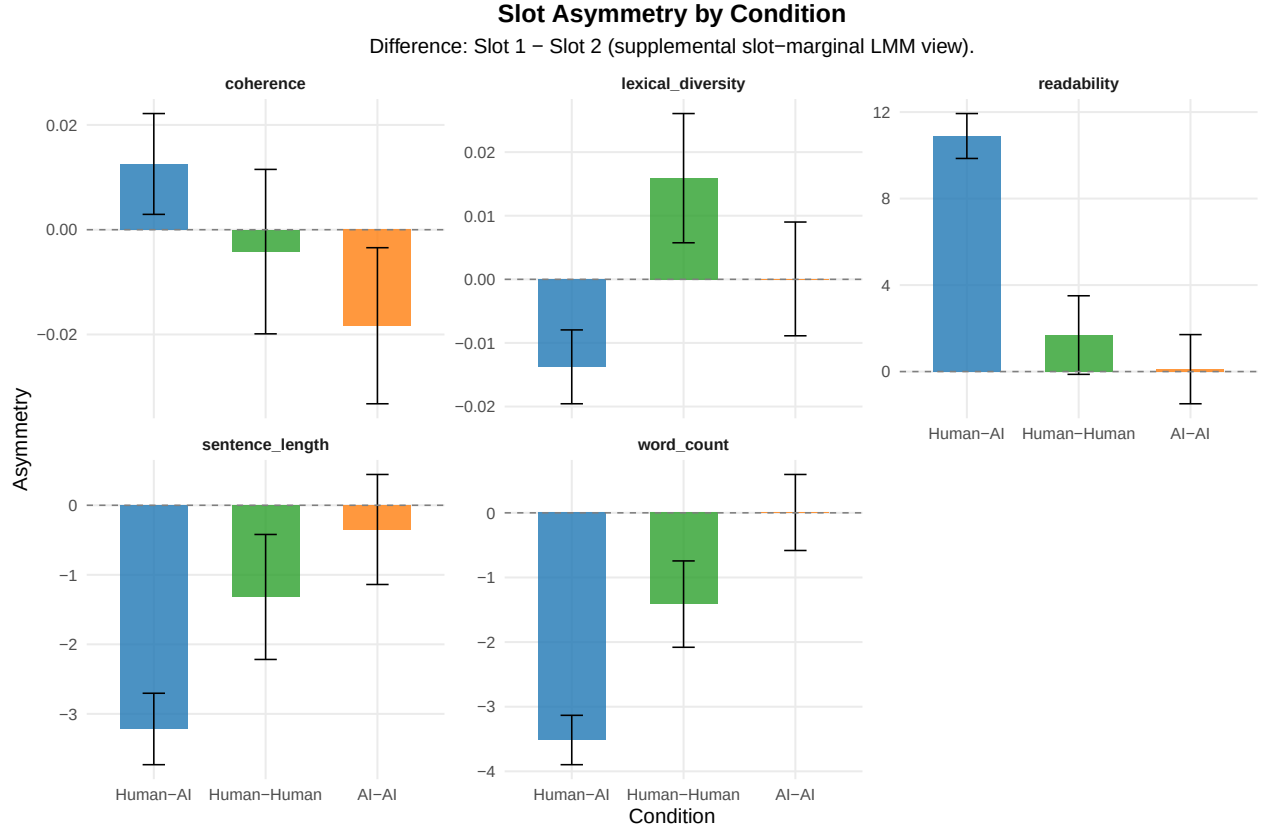


5 Supplemental: Slot-Marginal LMM per Metric

Table 5: Slot asymmetry by condition for selected surface metrics

metric	condition	slot_1_mean	slot_2_mean	difference	se	t	p
word_count	Human-AI	25.581	29.096	-3.515	0.195	-18.039	0.000
word_count	Human-Human	25.975	27.387	-1.412	0.341	-4.142	0.000
word_count	AI-AI	27.972	27.967	0.006	0.300	0.019	0.985
sentence_length	Human-AI	11.577	14.794	-3.217	0.262	-12.274	0.000
sentence_length	Human-Human	10.000	11.319	-1.319	0.459	-2.875	0.004
sentence_length	AI-AI	14.904	15.252	-0.348	0.404	-0.862	0.389
lexical_diversity	Human-AI	0.871	0.885	-0.014	0.003	-4.645	0.000
lexical_diversity	Human-Human	0.873	0.857	0.016	0.005	3.069	0.002
lexical_diversity	AI-AI	0.879	0.879	0.000	0.005	0.015	0.988
readability	Human-AI	90.727	79.835	10.892	0.530	20.560	0.000

metric	condition	slot_1_mean	slot_2_mean	difference	se	t	p
readability	Human-Human	94.081	92.393	1.688	0.927	1.821	0.069
readability	AI-AI	83.061	82.951	0.110	0.816	0.135	0.892
coherence	Human-AI	0.803	0.790	0.013	0.005	2.556	0.011
coherence	Human-Human	0.816	0.820	-0.004	0.008	-0.522	0.601
coherence	AI-AI	0.777	0.795	-0.018	0.008	-2.412	0.016



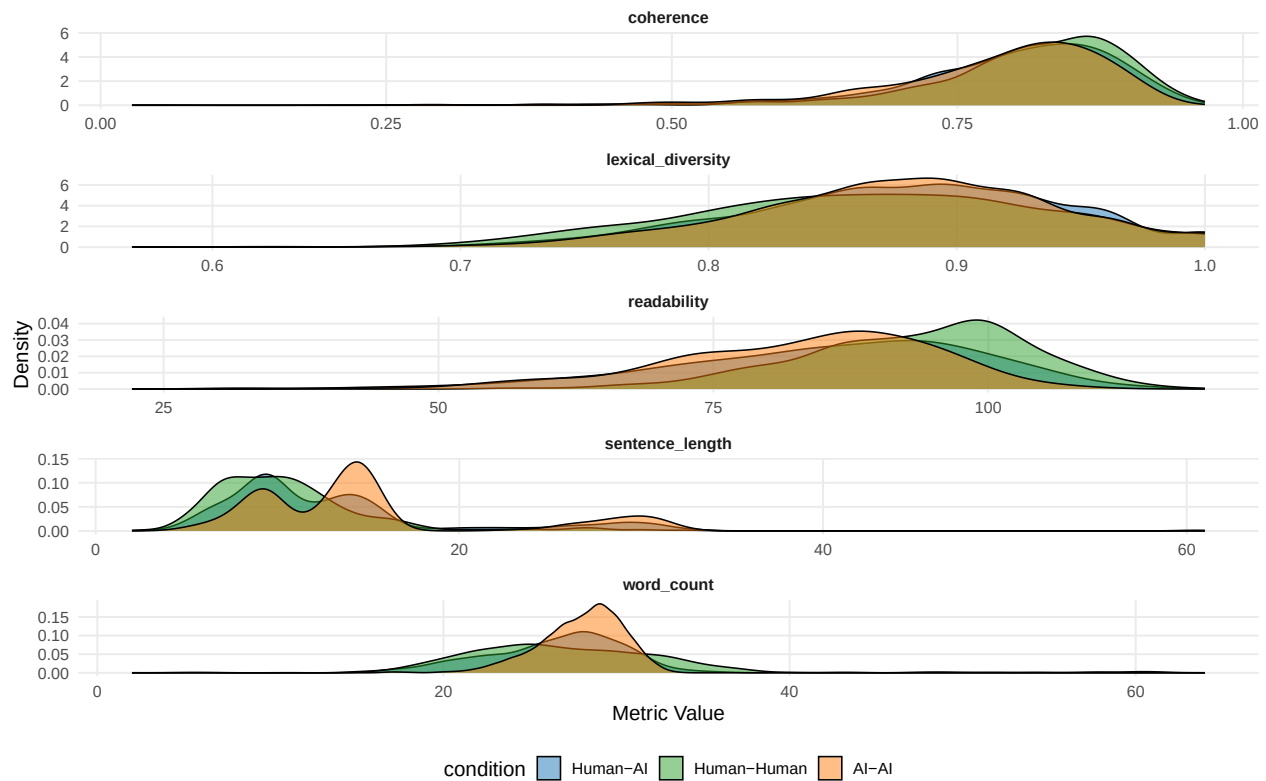
6 Supplemental: General Condition-Level Comparison

Table 6: Condition-level descriptives for selected surface metrics

condition	metric	mean_value	sd_value
Human-AI	word_count	27.320	6.430
Human-Human	word_count	26.681	4.848
AI-AI	word_count	27.969	2.409
Human-AI	sentence_length	13.231	7.108
Human-Human	sentence_length	10.660	4.521
AI-AI	sentence_length	15.078	6.987
Human-AI	lexical_diversity	0.878	0.065
Human-Human	lexical_diversity	0.865	0.071
AI-AI	lexical_diversity	0.879	0.061
Human-AI	readability	85.200	14.454

condition	metric	mean_value	sd_value
Human-Human	readability	93.237	10.314
AI-AI	readability	83.006	12.020
Human-AI	coherence	0.797	0.101
Human-Human	coherence	0.818	0.079
AI-AI	coherence	0.785	0.099

Surface Metric Distributions by Condition



```
##
## === word_count ( n_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 17561.2
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -6.8556 -0.5096  0.0784  0.5230  5.4445
##
## Random effects:
## Groups      Name                Variance Std.Dev.
## conversation_id (Intercept) 12.06      3.472
## Residual                18.17      4.263
## Number of obs: 2986, groups: conversation_id, 171
##
```

```

## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      27.3384    0.3623 167.1224  75.465  <2e-16 ***
## conditionHuman-Human -0.6574    0.7434 166.0012  -0.884   0.378
## conditionAI-AI       0.6310    0.6767 166.0732   0.933   0.352
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) cndH-H
## cndtnHmn-Hm -0.487
## condtnAI-AI -0.535  0.261
## contrast      estimate    SE df t.ratio p.value
## (Human-AI) - (Human-Human)   0.657 0.743 167   0.884  1.0000
## (Human-AI) - (AI-AI)        -0.631 0.677 168  -0.933  1.0000
## (Human-Human) - (AI-AI)     -1.288 0.865 167  -1.490  0.4146
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === sentence_length ( sentence_length_mean ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 19088.4
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.0017 -0.6012 -0.1422  0.3943  5.7626
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 13.60   3.688
## Residual              30.96   5.564
## Number of obs: 2986, groups: conversation_id, 171
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      13.1856    0.3927 169.4348  33.579  < 2e-16 ***
## conditionHuman-Human -2.5260    0.8052 167.7973  -3.137   0.00202 **
## conditionAI-AI       1.8924    0.7330 167.9023   2.582   0.01068 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr) cndH-H
## cndtnHmn-Hm -0.488
## condtnAI-AI -0.536  0.261
## contrast      estimate    SE df t.ratio p.value
## (Human-AI) - (Human-Human)   2.53 0.805 167   3.137  0.0061
## (Human-AI) - (AI-AI)        -1.89 0.733 167  -2.582  0.0321
## (Human-Human) - (AI-AI)     -4.42 0.937 167  -4.717  <.0001

```

```

##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === lexical_diversity ( proportion_unique_tokens ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: -7950.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.6028 -0.6301  0.0494  0.6772  2.6187
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0004956 0.02226
## Residual                    0.0037836 0.06151
## Number of obs: 2986, groups: conversation_id, 171
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    0.878084   0.002679 170.403592 327.757  <2e-16 ***
## conditionHuman-Human -0.012938  0.005472 166.211721  -2.364  0.0192 *
## conditionAI-AI      0.001311  0.004982 166.479791   0.263  0.7927
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.490
## condtnAI-AI -0.538  0.263
## contrast              estimate      SE df t.ratio p.value
## (Human-AI) - (Human-Human)  0.01294 0.00547 166   2.364  0.0577
## (Human-AI) - (AI-AI)       -0.00131 0.00498 166  -0.263  1.0000
## (Human-Human) - (AI-AI)    -0.01425 0.00636 165  -2.241  0.0790
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === readability ( flesch_reading_ease ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: 23472.8
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.5539 -0.6015  0.0999  0.6781  2.8232
##

```

```

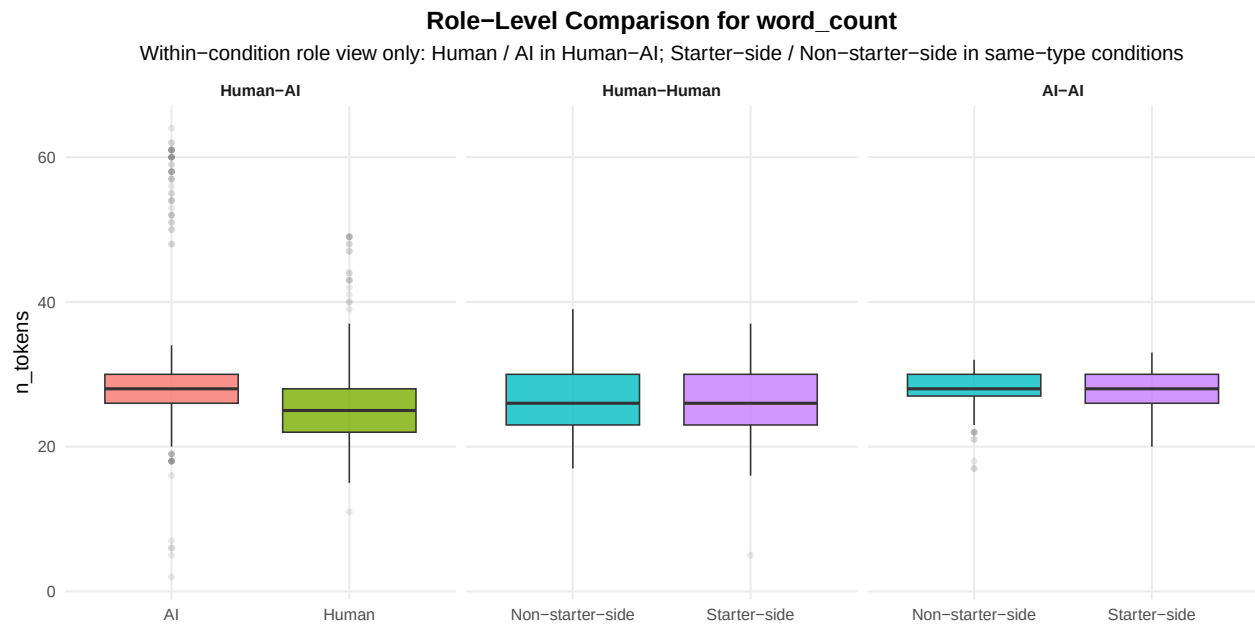
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 37.08    6.089
## Residual              137.86   11.741
## Number of obs: 2986, groups: conversation_id, 171
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      85.2791    0.6721 169.1973 126.881 < 2e-16 ***
## conditionHuman-Human  7.9578    1.3765 166.7362   5.781 3.57e-08 ***
## conditionAI-AI      -2.2733    1.2531 166.8939  -1.814  0.0714 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H
## cndtnHmn-Hm -0.488
## condtnAI-AI -0.536  0.262
## contrast              estimate    SE  df t.ratio p.value
## (Human-AI) - (Human-Human)   -7.96 1.38 167  -5.781 <.0001
## (Human-AI) - (AI-AI)         2.27 1.25 167   1.814  0.2143
## (Human-Human) - (AI-AI)      10.23 1.60 166   6.392 <.0001
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
##
## === coherence ( first_order_coherence ) ===
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: as.formula(paste(col_name, "~ condition + (1 | conversation_id)"))
## Data: df_combined
##
## REML criterion at convergence: -4772
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -7.7407 -0.4503  0.1633  0.6457  2.5032
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept) 0.0008302 0.02881
## Residual              0.0084964 0.09218
## Number of obs: 2563, groups: conversation_id, 171
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      0.797104    0.003792 169.796913 210.223 < 2e-16 ***
## conditionHuman-Human  0.020782    0.007564 153.275785   2.747  0.00673 **
## conditionAI-AI      -0.011409    0.007136 166.095666  -1.599  0.11179
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) cndH-H

```



```
## cndtnHmn-Hm -0.501
## condtnAI-AI -0.531 0.266
## contrast estimate SE df t.ratio p.value
## (Human-AI) - (Human-Human) -0.0208 0.00756 154 -2.747 0.0202
## (Human-AI) - (AI-AI) 0.0114 0.00714 167 1.598 0.3356
## (Human-Human) - (AI-AI) 0.0322 0.00891 156 3.612 0.0012
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: bonferroni method for 3 tests
```

7 Supplemental: Role-Level Comparison



```
##
## === Role contrast for word_count ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: 10299
##
## Scaled residuals:
## Min 1Q Median 3Q Max
## -6.1037 -0.5035 0.0205 0.4846 4.6698
##
## Random effects:
## Groups Name Variance Std.Dev.
## conversation_id (Intercept) 18.50 4.302
## Residual 20.68 4.548
## Number of obs: 1708, groups: conversation_id, 100
```

```

##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      29.0964    0.4575  110.9134   63.60  <2e-16 ***
## condition_roleHuman -3.5152    0.2201 1606.4692  -15.97  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlHmn -0.241
## contrast      estimate    SE   df t.ratio p.value
## AI - Human      3.52 0.22 1607  15.972  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: 3260.3
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.2688 -0.7333 -0.0420  0.7141  2.6563
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## conversation_id (Intercept)  5.408   2.326
## Residual                18.288   4.276
## Number of obs: 558, groups: conversation_id, 31
##
## Fixed effects:
##               Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      26.7742    0.4899  40.1823   54.652  <2e-16 ***
## condition_roleStarter-side -0.1864    0.3621 526.0000  -0.515    0.607
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.370
## contrast              estimate    SE   df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.186 0.362 526   0.515  0.6069
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:

```

```

## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: 3191.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.9283 -0.5796  0.1691  0.6762  2.4627
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 1.419      1.191
## Residual                4.424      2.103
## Number of obs: 720, groups: conversation_id, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      28.00278    0.21854  51.30301 128.136 <2e-16 ***
## condition_roleStarter-side -0.06667    0.15678 679.00000  -0.425  0.671
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.359
## contrast              estimate      SE df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.0667 0.157 679  0.425  0.6708
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for sentence_length ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
## Data: df_subset
##
## REML criterion at convergence: 11082.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5045 -0.6047 -0.1818  0.3729  5.2410
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 13.64      3.693
## Residual                34.16      5.845
## Number of obs: 1708, groups: conversation_id, 100
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      14.7946    0.4200 126.5688  35.22 <2e-16 ***

```

```

## condition_roleHuman   -3.2169      0.2829 1607.6182  -11.37   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## cndtn_rlHmn -0.337
## contrast      estimate      SE    df t.ratio p.value
## AI - Human      3.22 0.283 1607  11.373  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: 3258.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -1.6805 -0.6623 -0.1925  0.3109  5.2171
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept)  0.7956  0.892
## Residual                  19.5527  4.422
## Number of obs: 558, groups: conversation_id, 31
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      11.0514    0.3094  73.2373  35.715   <2e-16 ***
## condition_roleStarter-side -0.7836    0.3744 526.0000  -2.093   0.0368 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## cndtn_rlSt- -0.605
## contrast              estimate      SE    df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.784 0.374 526   2.093  0.0368
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: 4487.5

```

```

##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.2291 -0.7053  0.0685  0.5080  3.4982
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 23.70         4.868
## Residual                    25.52         5.052
## Number of obs: 720, groups: conversation_id, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      15.5219     0.8145  43.5195  19.057   <2e-16 ***
## condition_roleStarter-side -0.8880     0.3766  679.0000  -2.358   0.0186 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.231
## contrast                estimate      SE df t.ratio p.value
## (Non-starter-side) - (Starter-side)  0.888 0.377 679   2.358  0.0186
##
## Degrees-of-freedom method: kenward-roger
##
## === Role contrast for lexical_diversity ===
##
## Human-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##      Data: df_subset
##
## REML criterion at convergence: -4553.9
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.9736 -0.6207  0.0545  0.6904  2.5734
##
## Random effects:
##      Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0004195 0.02048
## Residual                    0.0037849 0.06152
## Number of obs: 1708, groups: conversation_id, 100
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      8.850e-01  2.938e-03  1.774e+02 301.201 < 2e-16 ***
## condition_roleHuman -1.376e-02  2.977e-03  1.607e+03  -4.621 4.12e-06 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##

```

```

## Correlation of Fixed Effects:
##           (Intr)
## cndtn_rlHmn -0.507
## contrast      estimate      SE    df t.ratio p.value
## AI - Human    0.0138 0.00298 1607   4.621  <.0001
##
## Degrees-of-freedom method: kenward-roger
##
## Human-Human
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: -1366.7
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -4.1590 -0.6518  0.0436  0.7255  2.1796
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0004418 0.02102
## Residual                  0.0046542 0.06822
## Number of obs: 558, groups: conversation_id, 31
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      0.866006   0.005562  55.807606 155.710  <2e-16
## condition_roleStarter-side -0.001720   0.005776 526.000000  -0.298   0.766
##
## (Intercept)          ***
## condition_roleStarter-side
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##           (Intr)
## cndtn_rlSt- -0.519
## contrast              estimate      SE    df t.ratio p.value
## (Non-starter-side) - (Starter-side) 0.00172 0.00578 526   0.298  0.7660
##
## Degrees-of-freedom method: kenward-roger
##
## AI-AI
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula:
## as.formula(paste(col_name, "~ condition_role + (1 | conversation_id)"))
##   Data: df_subset
##
## REML criterion at convergence: -2055.2
##

```

```

## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.96440 -0.64985  0.03278  0.63825  2.75075
##
## Random effects:
##   Groups             Name             Variance Std.Dev.
## conversation_id (Intercept) 0.0007296 0.02701
## Residual                    0.0030035 0.05480
## Number of obs: 720, groups: conversation_id, 40
##
## Fixed effects:
##
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      8.785e-01  5.156e-03 5.476e+01  170.38  <2e-16 ***
## condition_roleStarter-side 1.798e-03  4.085e-03 6.790e+02   0.44    0.66
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr)
## cndtn_rlSt- -0.396
## contrast                estimate      SE df t.ratio p.value
## (Non-starter-side) - (Starter-side) -0.0018 0.00408 679  -0.440  0.6599
##
## Degrees-of-freedom method: kenward-roger

```

8 Summary

Primary cross-condition result: per-metric $A_m(i) = |\overline{m}_{\text{slot } 1, i} - \overline{m}_{\text{slot } 2, i}|$, tested with ANOVA + Kruskal and BH-adjusted across metrics.

Supplemental views retained:

- LMM slot-marginal contrasts per metric (`metric ~ slot * condition + (1|story)`)
- condition-level distributions and marginal condition LMMs
- within-condition role comparisons (Human / AI in Human-AI; Starter-side / Non-starter-side in same-type)

Deferred to a follow-up pass:

- starter-mechanism analysis (signed, starter-normalized side differences across all conditions)
- Human-AI decomposition (`speaker_type \u00d7 speaker_is_starter` within Human-AI only)

Analysis completed: 2026-04-26 15:53:39.991723